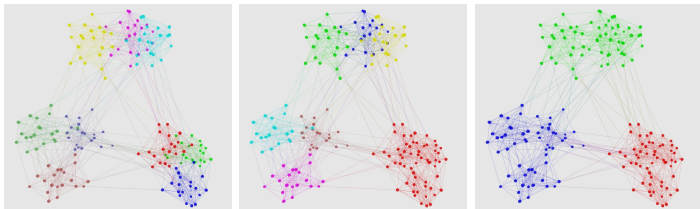


User Guide

Navigate through a graph clustered
with several scales of description

May, 06, 2026



Presentation

Networks are one of the main conceptual structures for modeling complex systems, where nodes represent the basic lowest-level entities of the system and edges represent their interactions. Graph clustering algorithms are then used to identify modules : over-densely connected sets of nodes of these networks as representations of entities at higher scales of description in these systems.

bec is a C++ graph clustering program that finds the entities of a network with a hand on the **description scales** within the **nPnB framework**.

■ **nPnB** : B. GAUME, I. ACHITOUV, AND D. CHAVALARIAS : *Two antagonistic objectives for one multi-scale graph clustering framework* Nature Scientific Reports, 15 (2025), p. 13368. <https://hal.science/hal-05046050v1>

■ **bec** : B. GAUME : *Find Optimal clusterings in nPnB framework is hard, BEC.2 : an algorithm to find high-performing ones* Forthcoming (2026)

■ **description scales** : B. GAUME : *Defining the appropriate scales for describing a complex system* Forthcoming (2026)